

EEG Foundation Models for Early Neurological Prognosis After Cardiac Arrest

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Abstract—Early neurological prognostication after cardiac arrest remains a significant clinical challenge. Electroencephalography (EEG) provides continuous, noninvasive monitoring of brain activity and offers valuable prognostic information. This study evaluates four pretrained EEG foundation models (LaBraM, BENDR, EEGPT, and CBraMod) for neurological outcome prediction using the multicenter I-CARE v2.1 database. EEG recordings acquired between 24 and 32 hours after cardiac arrest were used to fine-tune each model for binary prediction of Cerebral Performance Category (CPC 1–2 vs. CPC 3–5). A patient-level stratified 75/25 train-validation split with three-fold cross-validation was adopted to prevent data leakage. Performance was assessed using AUC, Cohen’s kappa, F1-score, recall, and specificity. LaBraM and EEGPT achieved the best performance, reaching an AUC of 1.00 in the latest monitoring windows and consistently outperforming BENDR and CBraMod. These results highlight the potential of EEG foundation models for automated early neurological prognostication after cardiac arrest.

I. INTRODUCTION

Cardiac arrest constitutes a medical emergency with an approximate in hospital mortality rate of 75%, and nearly 20–40% of survivors develop significant neurological disability that limits long-term functional independence [1], [2]. Hypoxic-ischemic brain injury caused by the interruption of cerebral blood flow frequently results in post-anoxic coma, making early and accurate neurological prognostication essential for guiding clinical decisions regarding patient management and life-sustaining therapies [3], [4]. Current international guidelines recommend a multimodal prognostic strategy that combines clinical examination, electrophysiological tests, and neuroimaging; however, reliable prediction during the early post-arrest period remains challenging because of inter-patient variability and the limited sensitivity of existing methods [5].

Electroencephalography (EEG) is widely used for continuous, noninvasive monitoring of brain activity in comatose patients after cardiac arrest, although its interpretation is often limited by observer variability and the complexity of prolonged recordings [3], [6]. To overcome these limitations,

quantitative EEG approaches have been developed to extract objective biomarkers from brain signals. Nevertheless, most existing methods rely on handcrafted features that provide static representations of the EEG and may not adequately capture the temporal dynamics associated with neurological recovery, motivating the development of deep learning models capable of learning more informative representations directly from EEG recordings [7].

Recently, EEG foundation models pretrained on large-scale datasets have demonstrated remarkable representation learning capabilities across diverse EEG analysis tasks. However, their ability to transfer this knowledge to neurological outcome prediction after cardiac arrest has not yet been systematically evaluated.

To address this gap, this paper investigates the adaptation of four pretrained EEG foundation models to predict the neurological prognosis of comatose patients after cardiac arrest using EEG recordings acquired between 24 and 32 hours following the event. The evaluated models include LaBraM, BENDR, EEGPT and CBraMod [8], [9], [10], [11]. These foundation models were pre-trained on large-scale EEG datasets collected from Brain Computer Interface (BCI) studies, comprising recordings from thousands of subjects across multiple institutions and a wide range of experimental protocols and clinical conditions. Their extensive pre-training enables the models to learn robust and generalizable EEG representations that can be adapted to downstream clinical tasks, such as neurological outcome prediction after cardiac arrest.

Recent studies have explored machine learning for neurological outcome prediction after cardiac arrest using quantitative EEG features, heart rate variability, and deep learning architectures such as CNN–LSTM models [12], [6]. Our previous work investigated a CNN–LSTM framework on the multicenter I-CARE v2.1 dataset, demonstrating the feasibility of deep learning for automated neurological prognostication from EEG recordings [13].

In contrast, this work investigates whether pretrained EEG

foundation models can transfer knowledge learned from large-scale EEG datasets to neurological outcome prediction after cardiac arrest.

II. METHODOLOGY

A. Dataset

The experiments were conducted using the multicenter I-CARE v2.1 database hosted on PhysioNet [14], [15], which contains continuous EEG recordings from 607 adult comatose patients following cardiac arrest. Data were collected across multiple intensive care units, resulting in heterogeneous acquisition protocols, including differences in sampling frequencies, channel configurations, and recording durations.

Neurological outcome was formulated as a binary classification task according to the Cerebral Performance Category (CPC) scale, where CPC scores of 1–2 indicate a good neurological outcome and CPC scores of 3–5 indicate a poor neurological outcome.

Following current clinical recommendations for early neurological prognostication [16], EEG recordings acquired between 24 and 32 hours after cardiac arrest were analyzed. Two temporal groups were evaluated: 24–26–28 hours and 28–30–32 hours. Each group consisted of three EEG recordings acquired at two-hour intervals. For every recording, a single 90-second EEG segment was extracted between minutes 2 and 3.5 to minimize transient artifacts. Only patients with EEG recordings available for all time points within the corresponding temporal group were included in each experiment.

B. EEG Preprocessing

EEG signals were preprocessed following standard clinical and processing signal guidelines. First, the data were band-pass filtered between 0.5 and 45 Hz using a zero-phase FIR filter implemented in MNE-Python, and subsequently resampled to 128 Hz in accordance with the Nyquist-Shannon sampling theorem, ensuring that the sampling frequency satisfied $f_s \geq 2f_{max}$ and $f_{max} = 45Hz$ [17]. Line noise was attenuated using a multi-taper approach without relying on fixed filtering assumptions. A robust average reference was then iteratively estimated while excluding noisy channels, which were identified based on statistical and spectral criteria. Finally, RANSAC was applied to detect channels that deviated from expected signals using spatial predictions from neighboring electrodes [18]. To train the model, EEG signals were analyzed using the standard 10–20 system with 19 mono-polar channels.

C. EEG Foundation Models

Four pretrained EEG foundation models were evaluated in this study: LaBraM [8], BENDR [9], EEGPT [10], and CBraMod [11]. These models were selected because they represent complementary approaches for learning transferable EEG representations through large-scale pretraining. In particular, they employ different self-supervised learning strategies and architectural designs while providing publicly available

pretrained weights, making them suitable for transfer learning on heterogeneous ICU EEG recordings.

Table I summarizes the main characteristics of the evaluated EEG foundation models.

TABLE I: Comparison of the evaluated EEG foundation models.

Model	Backbone	Learning Strategy	Clinical Motivation
LaBraM	Transformer	Neural codebook pretraining	Robust to heterogeneous electrode configurations and recording protocols
BENDR	Transformer	Contrastive self-supervised learning	Learns general EEG representations directly from raw signals
EEGPT	Transformer	Spatio-temporal representation learning	Produces universal EEG embeddings transferable to downstream clinical tasks
CBraMod	Criss-cross Transformer	Self-supervised pretraining	Models long-range spatial and temporal EEG dependencies

1) *LaBraM*: LaBraM is a transformer-based EEG foundation model pretrained on large-scale EEG datasets using neural codebook representations. Its flexible spatial and temporal embeddings facilitate adaptation to different electrode configurations, making it suitable for heterogeneous clinical EEG recordings [8].

2) *BENDR*: BENDR is a self-supervised transformer that learns EEG representations directly from raw signals using contrastive learning. Its hardware-agnostic design enables robust transfer across different recording systems without manual feature engineering, facilitating its application to heterogeneous clinical datasets [9].

3) *EEGPT*: EEGPT learns universal EEG representations through spatio-temporal alignment during large-scale pretraining. The resulting embeddings are robust to acquisition variability and have demonstrated strong transferability across downstream EEG classification tasks [10].

4) *CBraMod*: CBraMod employs a criss-cross transformer architecture to jointly model spatial and temporal EEG dependencies. Its attention mechanism is designed to capture long-range temporal dynamics, making it suitable for continuous EEG analysis in critical care applications [11].

D. Foundation Model Adaptation

The four pretrained EEG foundation models were fine-tuned for binary neurological outcome prediction. For each model, the original classification layer was replaced with a new binary classification head to predict good (CPC 1–2) or poor (CPC 3–5) neurological outcomes.

$$\hat{y} = g_\phi(f_\theta(x)), \quad (1)$$

where f_θ denotes the pretrained EEG foundation encoder, g_ϕ is the task-specific binary classification head, x represents the input EEG segment, and $\hat{y}$ is the predicted neurological outcome.

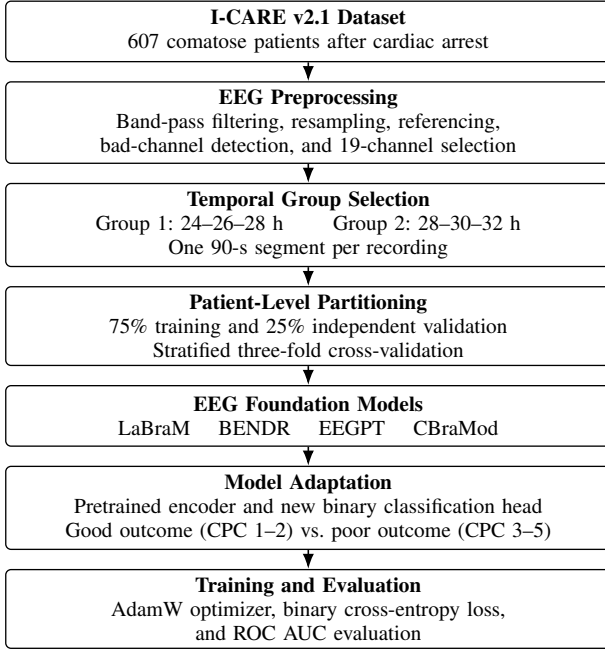


Fig. 1: Overview of the experimental workflow, including EEG preprocessing, temporal grouping, patient-level partitioning, foundation model adaptation, and performance evaluation.

The pretrained encoder was initialized using the publicly available weights provided by the original authors, while the new classification head was randomly initialized. Although each model requires a different input representation, all models were trained using the same data partitions, temporal groups, and evaluation protocol to ensure a fair comparison.

E. Training and Evaluation

All models were optimized using the AdamW optimizer and binary cross-entropy loss. Hyperparameters, including learning rate, batch size, and weight decay, were selected through cross-validation. Performance was primarily assessed using the area under the receiver operating characteristic curve (ROC AUC). All experiments were implemented in PyTorch and executed on an NVIDIA GPU.

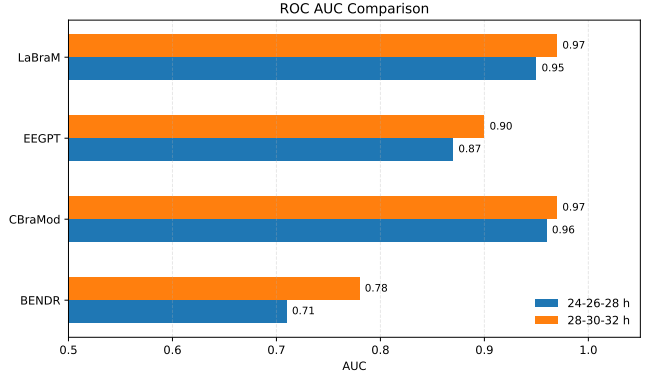
III. RESULTS AND DISCUSSION

Table II reports the numerical results, while Fig. 2 provides a visual comparison of model discrimination and classification agreement across the evaluated temporal groups.

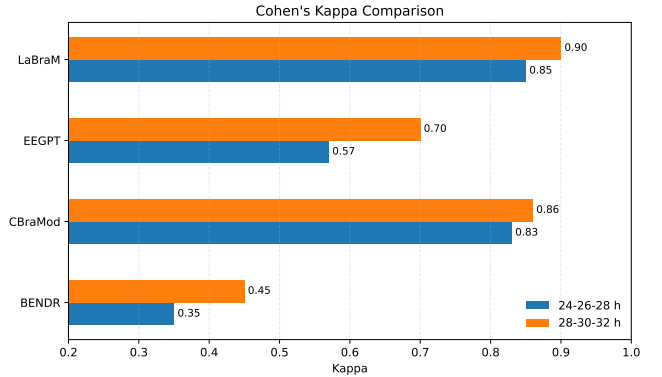
The evaluated EEG foundation models demonstrated strong predictive performance for neurological outcome prediction after cardiac arrest, with the highest results obtained using EEG recordings acquired between 28 and 32 hours after the event. LaBraM and CBraMod achieved the best discrimination, whereas LaBraM obtained the highest Cohen’s kappa coefficient, indicating superior agreement under class imbalance. EEGPT also showed competitive performance, while BENDR consistently achieved lower, although still acceptable, predictive accuracy. Models trained using EEG recordings from the earlier 24–28 hour interval exhibited only a moderate reduction

TABLE II: Performance comparison of different models with the best tuning features.

Model	Hours	Best tuning parameters: Learning rate, Max batch size	AUC score	Cohen Kappa score
Labram	24, 26, 28 h	2×10^{-4} , 16	0.95	0.85
	28, 30, 32 h	6×10^{-4} , 16	0.97	0.9
BENDR	24, 26, 28 h	6×10^{-6} , 32	0.71	0.35
	28, 30, 32 h	1×10^{-6} , 32	0.78	0.45
EEGPT	24, 26, 28 h	1×10^{-4} , 12	0.87	0.57
	28, 30, 32 h	6×10^{-4} , 12	0.9	0.7
CBraMod	24, 26, 28 h	1×10^{-5} , 32	0.96	0.83
	28, 30, 32 h	6×10^{-5} , 32	0.97	0.86



(a) ROC AUC comparison.



(b) Cohen’s kappa comparison.

Fig. 2: Comparison of the performance obtained by the evaluated EEG foundation models for the two post-cardiac arrest time windows. (a) ROC AUC values. (b) Cohen’s kappa coefficients. The later time window (28–30–32 h) consistently produced higher performance, with LaBraM and CBraMod achieving the best overall results.

in performance, indicating that clinically relevant prognostic information is already present within the first day after cardiac arrest.

LaBraM, CBraMod, and EEGPT consistently outperformed BENDR, suggesting that large-scale self-supervised pretraining substantially improves downstream neurological outcome prediction. Their superior performance is likely attributable to the diversity and scale of the datasets used during pretraining, including variability in recording tasks, subjects, acquisition conditions, and temporal characteristics, which enable the

models to learn highly transferable EEG representations [19]. In contrast, although BENDR remained competitive, its lower performance may reflect differences in its pretraining strategy and representational capacity relative to the more recent foundation models.

The results demonstrate that raw EEG signals alone contain sufficient information to predict neurological outcomes within the first 24–32 hours following cardiac arrest. Although predictive performance improved when later EEG recordings were used, reliable predictions were already achievable using recordings acquired during the first 24 hours. This finding is clinically important because earlier prognostication may facilitate timely therapeutic decision-making and optimize patient management. Moreover, foundation models trained on heterogeneous EEG datasets are expected to provide greater robustness across different recording conditions, increasing their potential applicability in intensive care environments.

Previous studies have reported excellent performance by combining multiple clinical modalities. For example, Niu et al. achieved an AUC of 1.00 using a multimodal approach that integrated EEG, ECG, and demographic variables only 12 hours after cardiac arrest [12]. However, the PhysioNet I-CARE database contains missing values in several demographic variables, particularly Shockable Rhythm, one of the strongest predictors of neurological outcome [14]. In contrast, the proposed approach relied exclusively on raw EEG recordings and achieved comparable predictive performance using recordings acquired within the first 24 hours after cardiac arrest. These results suggest that EEG alone contains sufficiently informative neurophysiological patterns for accurate prognostication while avoiding dependence on incomplete clinical variables. Furthermore, unlike conventional machine learning or CNN–LSTM approaches, foundation models eliminate the need for handcrafted feature extraction by directly learning robust representations from raw EEG signals.

Despite these promising results, several limitations should be acknowledged. First, the study was conducted using a single retrospective dataset, and external validation on independent cohorts from different institutions is still required to assess model generalizability. Second, the high AUC values achieved by LaBraM and CBraMod should be interpreted cautiously, as they require confirmation on external datasets to exclude potential overfitting or dataset-specific biases. Finally, although the evaluated foundation models achieved excellent predictive performance, they remain largely non-interpretable, limiting their clinical adoption because transparent and explainable clinical decision-making is essential in critical care.

Future work will focus on evaluating the generalizability of the best-performing models through cross-hospital validation, in which training and testing are performed using data collected from different medical centers. Additional studies will investigate whether comparable predictive performance can be achieved using EEG recordings acquired at earlier post-cardiac arrest time windows, enabling even earlier neurological prognostication. Furthermore, incorporating explainability techniques will be essential for identifying the neurophysiological patterns driving model predictions and increasing clinicians' confidence in these systems. Finally, prospective

clinical validation will be necessary to determine the safety, robustness, and practical utility of EEG foundation models before their integration into routine intensive care practice.

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